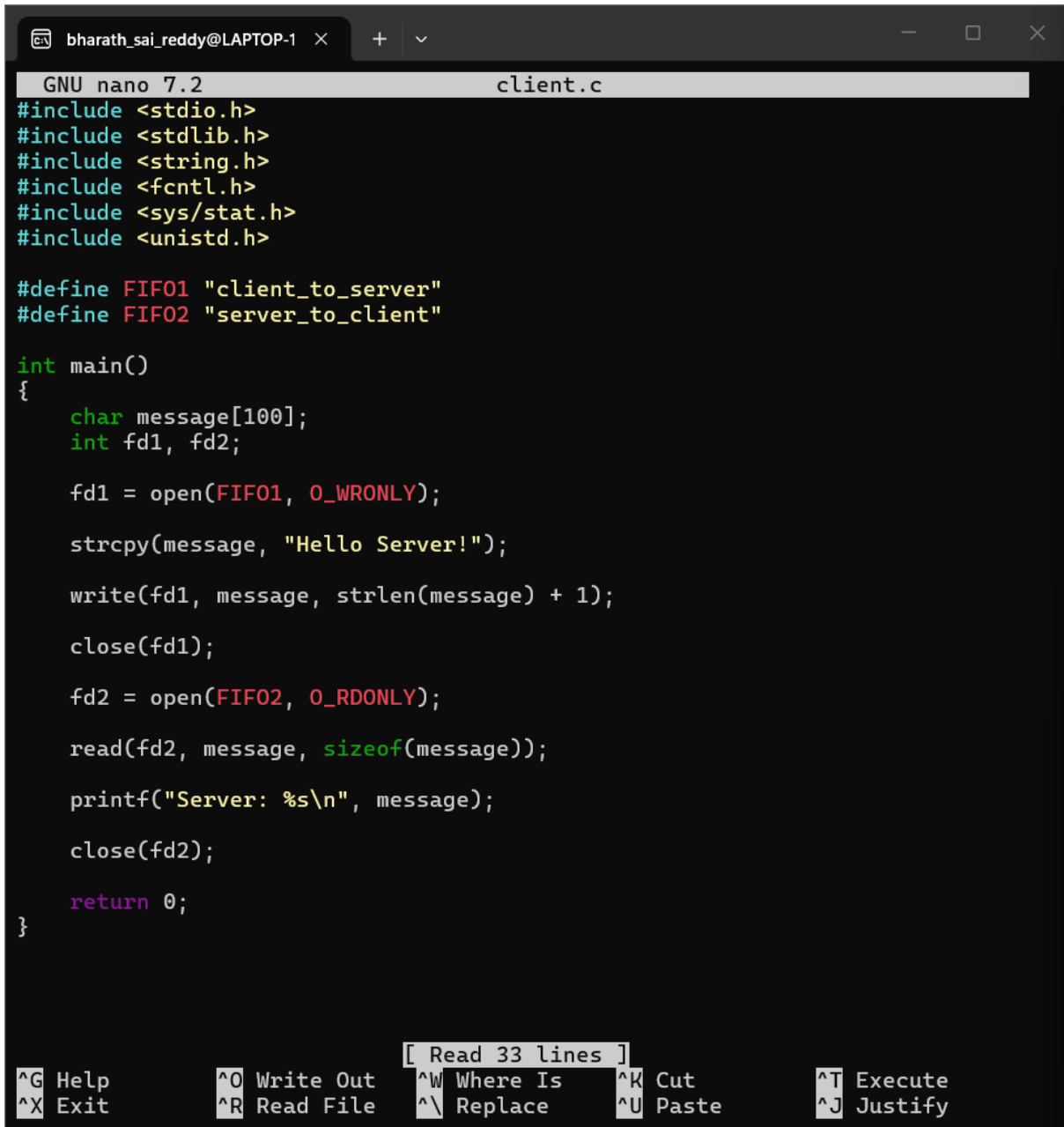


Practical -06

Code:

Client code:



The screenshot shows a terminal window with the nano 7.2 text editor open. The file being edited is client.c. The code is a C program for a client that sends a message to a server and receives a response. The code includes standard headers, defines FIFOs, and uses open, write, read, and close system calls. The terminal window has a dark background with syntax highlighting. The bottom of the window shows a status bar with various keyboard shortcuts.

```
GNU nano 7.2 client.c
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <fcntl.h>
#include <sys/stat.h>
#include <unistd.h>

#define FIF01 "client_to_server"
#define FIF02 "server_to_client"

int main()
{
    char message[100];
    int fd1, fd2;

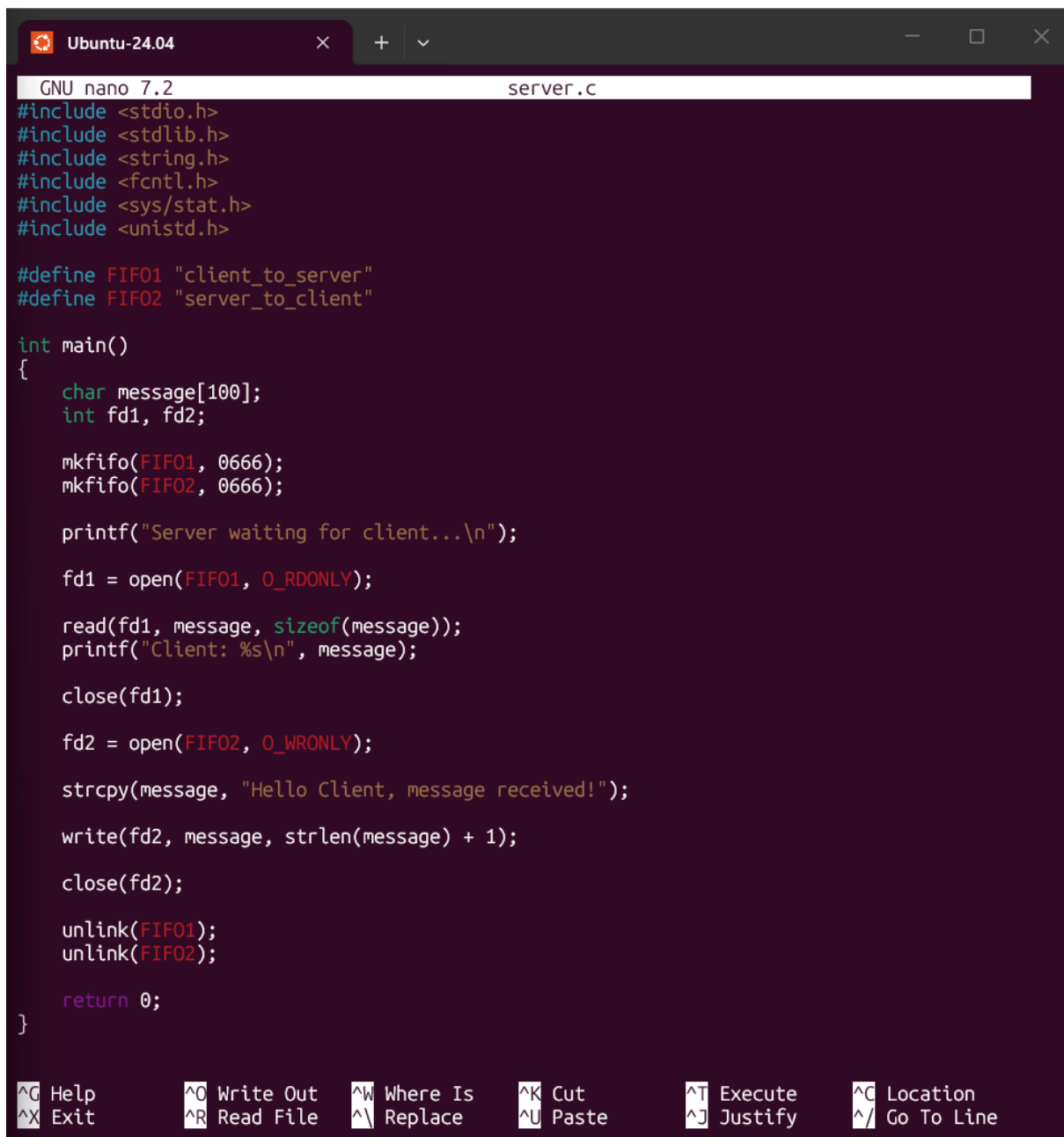
    fd1 = open(FIF01, O_WRONLY);
    strcpy(message, "Hello Server!");
    write(fd1, message, strlen(message) + 1);
    close(fd1);

    fd2 = open(FIF02, O_RDONLY);
    read(fd2, message, sizeof(message));
    printf("Server: %s\n", message);
    close(fd2);

    return 0;
}
```

^G Help ^O Write Out [Read 33 lines] ^W Where Is ^K Cut
^X Exit ^R Read File ^\ Replace ^U Paste ^T Execute
^J Justify

Server Code:



```
GNU nano 7.2 server.c
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <fcntl.h>
#include <sys/stat.h>
#include <unistd.h>

#define FIFO1 "client_to_server"
#define FIFO2 "server_to_client"

int main()
{
    char message[100];
    int fd1, fd2;

    mkfifo(FIFO1, 0666);
    mkfifo(FIFO2, 0666);

    printf("Server waiting for client...\n");

    fd1 = open(FIFO1, O_RDONLY);

    read(fd1, message, sizeof(message));
    printf("Client: %s\n", message);

    close(fd1);

    fd2 = open(FIFO2, O_WRONLY);

    strcpy(message, "Hello Client, message received!");

    write(fd2, message, strlen(message) + 1);

    close(fd2);

    unlink(FIFO1);
    unlink(FIFO2);

    return 0;
}
```

Help Exit Write Out Read File Where Is Replace Cut Paste Execute Justify Location Go To Line

OutPut:

```
bharath_sai_reddy@LAPTOP-1  x  +  v  -  □  x
System load:  0.56      Processes:           67
Usage of /:   0.3% of 1006.85GB  Users logged in:      0
Memory usage: 6%      IPv4 address for eth0: 172.29.142.237
Swap usage:   0%

* Canonical Workshop gives developers fast, composable, reproducible, and
secure developer environments that are perfect for agentic workflows.

https://ubuntu.com/workshop

This message is shown once a day. To disable it please create the
/home/bharath_sai_reddy/.hushlogin file.
bharath_sai_reddy@LAPTOP-1711DF88:~$ ls
90128_OSSP  OSLab  filedescriptor  markdown-project
bharath_sai_reddy@LAPTOP-1711DF88:~$ cd 90128_OSSP
bharath_sai_reddy@LAPTOP-1711DF88:~/90128_OSSP$ ls
2520090128_Practical  2520090128_Skill  File_Permission  ForgeOS  Scheduling
algorithms  filedescriptor  pipes  signals
bharath_sai_reddy@LAPTOP-1711DF88:~/90128_OSSP$ cd 2520090128_Practical
bharath_sai_reddy@LAPTOP-1711DF88:~/90128_OSSP/2520090128_Practical$ ls
Practical-01  Practical-03  Practical-05  Practical-07  Practical-09
Practical-02  Practical-04  Practical-06  Practical-08  Practical-10
bharath_sai_reddy@LAPTOP-1711DF88:~/90128_OSSP/2520090128_Practical$ cd Prac
tical-06
bharath_sai_reddy@LAPTOP-1711DF88:~/90128_OSSP/2520090128_Practical/Practica
l-06$ ls
bharath_sai_reddy@LAPTOP-1711DF88:~/90128_OSSP/2520090128_Practical/Practica
l-06$ nano server.c
bharath_sai_reddy@LAPTOP-1711DF88:~/90128_OSSP/2520090128_Practical/Practica
l-06$ nano client.c
bharath_sai_reddy@LAPTOP-1711DF88:~/90128_OSSP/2520090128_Practical/Practica
l-06$ gcc server.c -o server
gcc client.c -o client
bharath_sai_reddy@LAPTOP-1711DF88:~/90128_OSSP/2520090128_Practical/Practica
l-06$ ./server
Server waiting for client...
Client: Hello Server!
bharath_sai_reddy@LAPTOP-1711DF88:~/90128_OSSP/2520090128_Practical/Practica
l-06$ |

Ubuntu-24.04  x  +  v  -  □  x
bharath_sai_reddy@LAPTOP-1711DF88:~$ ls
90128_OSSP  OSLab  filedescriptor  markdown-project
bharath_sai_reddy@LAPTOP-1711DF88:~$ cd 90128_OSSP
bharath_sai_reddy@LAPTOP-1711DF88:~/90128_OSSP$ ls
2520090128_Practical  2520090128_Skill  File_Permission  Schedulingalgorithms  pipes
2520090128_Skill  ForgeOS  filedescriptor  signals
bharath_sai_reddy@LAPTOP-1711DF88:~/90128_OSSP$ cd 2520090128_Practical
bharath_sai_reddy@LAPTOP-1711DF88:~/90128_OSSP/2520090128_Practical$ ls
Practical-01  Practical-03  Practical-05  Practical-07  Practical-09
Practical-02  Practical-04  Practical-06  Practical-08  Practical-10
bharath_sai_reddy@LAPTOP-1711DF88:~/90128_OSSP/2520090128_Practical$ cd Prac
tical-06
bharath_sai_reddy@LAPTOP-1711DF88:~/90128_OSSP/2520090128_Practical/Practica
l-06$ ./client
Server: Hello Client, message received!
bharath_sai_reddy@LAPTOP-1711DF88:~/90128_OSSP/2520090128_Practical/Practica
l-06$
```